INCH-POUND

A-A-54735
10 September 1992

COMMERCIAL ITEM DESCRIPTION

STERILIZER, SURGICAL INSTRUMENT AND DRESSING

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for MIL-S-36500A, which is cancelled.

This Commercial Item Description covers one type of ultra-lightweight, portable sterilizer suitable for use in field operations.

Salient Characteristics:

The unit shall be an ultra-lightweight, portable, vertical, pressure sterilizer, externally heated and suitable for sterilizing surgical instruments and dressings by special warfare and associated indigenous personnel engaged in field operations.

The sterilizer shall provide steam sterilization of composite loads of dressings and surgical instruments for a minimum of thirty minutes at 250 degrees F. Dressings shall be dry and unscorched at the end of the sterilizing cycle.

The portable sterilizer shall be constructed of corrosion-resisting metal and shall meet the American Society of Mechanical Engineers' (ASME) code for Power Boilers.

Design shall permit total displacement of the air in the chamber by steam.

Size:

a. Minimum internal size of sterilizer chamber: 12-1/2 inch diameter by 12-1/4 inch height.

b. Maximum external dimensions for packing: 15 inches width by 16 inches length by 19 inches height.

AMSC N/A FSC 6530

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c. Maximum overall weight: 24 pounds.

The sterilizer shall be suitable for use with external heat sources ranging from electricity (hot plate) or a gasoline burner to open fires (coal, coke, wood).

The sterilizer shall be capable of being refilled with water and/or reloaded with material to be sterilized without complete cooling of the sterilizer.

Access to the sterilizer chamber shall be possible only after the pressure has dropped to a safe level (4 psig or less).

Two (2) copies of operating instructions shall be provided with the equipment. Instructions shall be laminated with clear plastic or provided in a waterproof container to protect them from water damage.

The sterilizer (autoclave) shall be capable of withstanding the following conditions:

a. Safe storage and transportation without permanent impairment of its capabilities from the effects of extreme conditions

b. Phase I airborne operations (parachute delivery)

c. Transportation in undeveloped areas (man-packed, and in military vehicles)

Design and construction shall be such as to permit operation and maintenance by unskilled personnel after limited training.

The sterilizer shall be capable of storage and transit under the following temperatures:

a. High temperature storage: Store unit at 155 degrees F for 3 days. After return to 70 degrees F operate for 4 hours and inspect for damage.

b. Extreme low temperature storage: Store unit at minus 80 degrees F for 3 days. After return to 70 degrees F operate for 4 hours and inspect for damage.

Workmanship. The sterilizer shall be free from defects which detract from its appearance or impair its serviceability.

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Unit. Each (EA). One sterilizer, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of III. The unit of product for examination purposes shall be one sterilizer.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of II. The unit of product for test purposes shall be one sterilizer.

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Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to, the following:

Operational test. The unit shall be loaded with the maximum possible number of sterilization test packs, filled with a 700 plus or minus 25 parts per million solution of sodium chloride (Na Cl) in water (brine), closed, and heated up with a controllable heat source to its maximum operating temperature for four minutes with the control valve open. The heat level shall then be maintained with the control valve closed and the internal pressure allowed to build up until the safety valve is activated. If the pressure gauge, which has previously been calibrated to react with plus or minus 2.5 percent of its full scale reading, indicates more than 26 PSI or less 21 PSI at this point, the test shall be stopped and the article immediately rejected. The heat source shall then be turned off and the unit allowed to cool off with the control valve re-opened. The unit shall be refilled with the brine solution and the above test repeated for a total of twenty (20) times. After the first trial, if the trip point of the safety valve should fall outside of the acceptable range, recalibration of the pressure gauge and cleaning, adjustment or replacement of the safety valve shall be allowed in order to permit the test to proceed. At the conclusion of the last test, there shall be no evidence of stress corrosion.

Notes: (1) The test packs need not be changed before each cycle since their purpose in this test is to provide a uniform reproducible load.

(2) Manufacturer is responsible for cleaning and drying the water pathway and all welded components in the unit. Care should be taken to insure that the sterilizer is free of all sodium chloride added for testing purposes.

Ruggedness test. The sterilizer shall be subjected to a transit drop test. This test requires the sterilizer to be dropped from 48 inches on each face, edge and corner. Drop shall be made from a quick-release hook, or drop tester. The floor or barrier receiving the impact shall be of two-inch plywood backed by concrete. There shall be no physical damage or impairment of the sterilizer operating efficiency at the conclusion of test.

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Performance test. The sterilizer shall be loaded with two fresh (never-used) sterilization test packs plus five (5) test strips. One test strip shall be placed in the center and one at each longitudinal end sandwiched between the two test packs. Indicator sealing tape shall be used to hold the test strips in place, if necessary, and seal the test packs together. The last two strips shall be placed on either side of the test packs inside the bottom of the chamber. Operating in accordance with the manufacturer's instructions, the sterilizer shall be secured and heated up to sterilization conditions where it shall be maintained for a period of at least 35 minutes before removing from heat source and allowing internal sterilizer pressure to drop to zero. The chamber shall then be opened and the spore strips removed in such a manner as to preclude contamination and tested for sterility in accordance with the U.S. Pharmacopeia. The weight gain of the test packs shall not exceed 3 percent of the pre-test weight. The ink on the sheets in the test pack shall have a uniform color.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

6530-00-926-2022

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MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA-MPQAS
USPHS
VA - OSS

Project Number: 6530-2347

Location: STERC1B.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I. RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-54735	2. DOCUMENT DATE (YYMMDD): 10 September 1992

3. DOCUMENT TITLE: STERILIZER, SURGICAL INSTRUMENT AND DRESSING

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870 (2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340, AUTOVON 289-2340

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